

Directions Toward Effective Utilization of Tactile Skin: A Review

Ravinder S. Dahiya, *Senior Member, IEEE*, Philipp Mittendorf, Maurizio Valle, *Member, IEEE*, Gordon Cheng, *Senior Member, IEEE*, and Vladimir J. Lumelsky, *Life Fellow, IEEE*

Abstract—A wide variety of tactile (touch) sensors exist today for robotics and related applications. They make use of various transduction methods, smart materials and engineered structures, complex electronics, and sophisticated data processing. While highly useful in themselves, effective utilization of tactile sensors in robotics applications has been slow to come and largely remains elusive today. This paper surveys the state of the art and the research issues in this area, with the emphasis on effective utilization of tactile sensors in robotic systems. One specific with the use of tactile sensing in robotics is that the sensors have to be spread along the robot body, the way the human skin is – thus dictating varied 3-D spatio-temporal requirements, decentralized and distributed control, and handling of multiple simultaneous tactile contacts. Satisfying these requirements pose challenges to making tactile sensor modality a reality. Overcoming these challenges requires dealing with issues such as sensors placement, electronic/mechanical hardware, methods to access and acquire signals, automatic calibration techniques, and algorithms to process and interpret sensing data in real time. We survey this field from a system perspective, recognizing the fact that the system performance tends to depend on how its various components are put together. It is hoped that the survey will be of use to practitioners designing tactile sensing hardware (whole-body or large-patch sensor coverage), and to researchers working on cognitive robotics involving tactile sensing.

Index Terms—Tactile skin, interface electronics, modular, embedded systems, sensor fusion, bendable electronics.

I. INTRODUCTION

RAPID technological advances and increasing expectations have been changing the face of robotics field in recent decades. The concept of a robot as an industrial tool operating in a “no-humans allowed” zone (but controlled

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R. S. Dahiya is with the Electronics and Nanoscale Engineering Research Division, University of Glasgow, Glasgow G128QQ, U.K. (e-mail: Ravinder.Dahiya@glasgow.ac.uk).

P. Mittendorf and G. Cheng are with the Institute for Cognitive Systems, Technische Universität München, Munich 80333, Germany (e-mail: philipp.mittendorf@tum.de; gordon@tum.de).

M. Valle is with DITEN, University of Genoa, Genoa 16145, Italy (e-mail: maurizio.valle@unige.it).

V. J. Lumelsky is with the University of Wisconsin-Madison, Madison, WI 53706 USA, and also with the University of Maryland, College Park, MD 20740 USA (e-mail: lumelsky@engr.wisc.edu).

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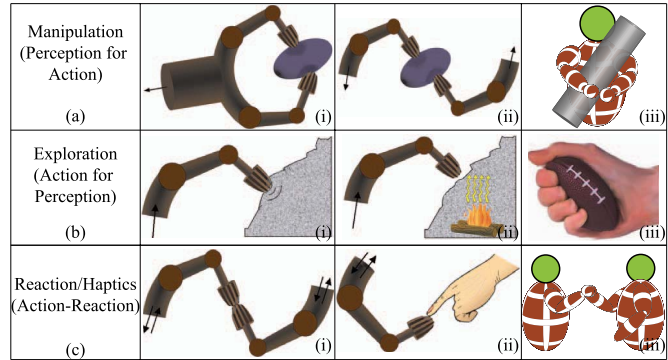


Fig. 1. Main robotic tasks involving tactile sensing (a) Object manipulation by a robot or multiple robots with sensors over smaller (i and ii) or large areas (iii). The data flows from object towards the controller; (b) Exploring objects to measure tactile features such as texture (i), temperature (ii), and softness/hardness (iii) etc. Here tactile data flows from controller towards the contact; (c) The reaction tasks where a robot or multiple robots react to a stimuli applied at a single contact point (i and ii) or over large area (iii). The action–reaction cycle involves bi-directional flow of signal [8].

by humans) is being augmented with new generation robots such as humanoids, social robots, flying robots, assistive and medical robots etc. making way in-to our lives. The trend is to make these robots operate autonomously; interact safely (causing no harm to themselves, to humans and to other surrounding objects); possess qualities like self maintenance, and more. Attaining these features is easier if a complete model of the environment is available, and if the robot actuators execute motion commands perfectly relative to this model. Unfortunately, a complete world model is not available in most cases of interest, and robots have to plan and successfully execute the tasks in presence of environmental uncertainties. This makes sensing an important component of robotics.

Among various sensory modalities, the tactile (touch) sensory modality is particularly important since most tasks (Fig. 1) require the robot to make a physical contact with objects. Tactile sensing plays a fundamental role in providing: (a) action–related information such as slip detection [1]; (b) control parameters in manipulation and control tasks such as grasping [2]–[4]; and (c) an estimation of contact parameters such as contact force (normal and shear), soft contact, hardness, texture, temperature, and slip [5]–[7]. The above functions are important constituents for many robotic tasks including safe interaction, making tactile sensory modality vital for utilization of the next generation robotics.

TABLE I
CLASSIFICATION OF VARIOUS TACTILE SENSING TECHNOLOGIES

Transduction Medium	Sensor Material	Sensor Structure
Resistive [37]–[42]	Conductive Polymer Composites [12], [71]–[76]	Microelectromechanical systems [88]–[97]
Capacitive [5], [43]–[45]	Carbon Nano Tubes [77]–[79]	Plastic MEMS [98]–[100]
Optical [46]–[49]	Force Sensing Resistors [30], [38]	Silicon Transistors [84]–[86], [101]–[103]
Magnetic [50]–[53]	Conductive Gels [36], [80]	Extended Gate Transistors [82], [104]–[106]
Ultrasonic [54]–[58]	Conductive Fibers and Yarns [81]	Organic Field Effect Transistors [81], [107], [108]
Piezoelectric [59]–[64]	Piezo-/pyroelectric [82]–[86]	Flexible Printed Circuit Boards [44]–[46], [109], [110]
Electrorheological [65]–[67]	Electro-Optic Materials [87]	Mechanical Switches
Magnetorheological [68]–[70]		

A wide variety of tactile sensors capable of handling and measuring various contact parameters have been reported in literature. Some of these, given in Table I [8], are classified on the basis of (a) transduction methods, (b) smart/engineered materials that intrinsically convert mechanical stimulus into usable signals, and (c) engineered structures that generate a signal on touch. Many of these tactile sensors are capable of measuring a single contact parameter, and a few can measure more than one contact, as evident from various survey papers and books on tactile sensing [8]–[26]. Nevertheless, the existence of this wide spectrum of tactile sensors so far has not made tactile sensing an “effective” presence in robotics, as evident from the limited use of tactile sensors in existing robotic platforms (e.g. humanoid platforms such as iCub, ASIMO, Robonaut, HRP, or biomedical instruments). By “effective” we mean real involvement of tactile sensing data in carrying out the robot’s tasks. The above can be attributed to a host of challenges arising from features like non-centralized or distributed nature of tactile sensing, varied spatio-temporal requirements over the body or in 3D space, and multiple contact parameters to be measured. Unlike with vision cameras, the above attributes make it difficult to have a unique technological solution for tactile sensing.

Whole is more than the sum of its parts. The overall performance of a system that makes use of tactile sensors is not only dictated by how good its components are, but also how well they are integrated to achieve the goal. As an example, it is known that for the human sense of touch the large number of receptors alone is not sufficient to perceive the real world [27], [28]. A number of other components of the somatosensory system are involved in the perceptual mechanism. Taking cue from the human sense of touch, the “effective” utilization of tactile sensors requires research attention towards issues like sensing of curved surfaces, deciphering the information in tactile data etc. An holistic viewpoint would highlight many factors that significantly contribute to the effectiveness of tactile data utilization. These include: (a) sensor performance (e.g. sensitivity, ability to measure various contact parameters); (b) physical aspects (e.g. sensors placement, conformability, wiring); (c) data processing and hardware issues (data acquisition, signal conditioning, communication, power, compatibility with existing electronic and sensing hardware); (d) algorithms and software (processing data from tactile sensors distributed in 3D space, sensor representation, deciphering tactile information); (e) engineering issues (integration of sensing structures with robots, maintenance, reliability).

This paper surveys the state of tactile sensing in robotics with regard to aforementioned issues. The goal is to highlight the bottlenecks and consider alternatives. There have been other surveys that discussed issues with utilization of tactile sensing [13], [25], [29]. While some topics tackled in this text do overlap with those prior surveys, the focus of this paper is different – it concentrates on added topics, such as sensor distribution and placement, feature extraction, data representations and sensor survivability issues. As “tactile sensor” technologies have been surveyed extensively in the past [8]–[13], [16]–[26]: we limit ourselves to concise summary of various types of tactile sensors (see Table I). Some of these sensor technologies for robotic applications have reached commercial stage and some of the examples are: FSR (Force Sensing Resistor) commercially available from Interlink [30]; pressure sensitive ink or rubber based touch sensors available from Tekscan [31]; touch sensors based on Quantum Tunnel Composites (QTC) available from Peratech [32]; capacitive technology based touch sensors such as RoboTouch and Dig-iTacts from Pressure Profile Systems [33]; optical mode of transduction based sensors by KINOTEX [34]; MEMS based touch sensors by TactoLogic [35]; and, conductive fluid based sensors such as BioTac by SynTouch [36].

The paper is organized as follows: A historical perspective of robotic tactile sensing is given in Section II. This is followed by a survey of hardware and software issues related to effective utilization of tactile sensing. Section III surveys the field in terms of physical or mechanical issues such as conformability, sensor distribution, optimal sensor placement and routing of wires. Work on electrical and electronic hardware, interface electronics, communication and related areas is surveyed in Section IV. Research issues related to higher perceptual and processing levels e.g. data interpretation and sensor fusion are discussed in Section V. Some engineering issues are tackled in Section VI. A summary of the conclusions and future directions appears in Section VII.

II. HISTORICAL PERSPECTIVE

Touch sensing in robotics began to develop around 1970s – albeit at a slower pace compared with other sensory modalities like vision [16], [24]. In its early days, the use of touch sensing largely remained restricted to joint force/torques sensors or intrinsic/internal touch sensors. The term “intrinsic” here refers to the touch sensors that measure parameters like contact force or joint angle without physically coming in contact with the

objects. Such sensors are usually located within the body of a robot. From an operational viewpoint, the intrinsic touch and vision were considered sufficient for the industrial setting. For this reason, until the end of 1990s, sensor based robotics was biased towards using vision and intrinsic touch sensing. This is evident from a large number of articles on vision and joint force/torque sensors published during that period.

The tactile sensing began to develop in robotics in 1980s; early surveys show a wide variety of tactile sensors developed during that period [21], [111]. The tactile sensing may also be termed as extrinsic/external touch sensing, as it refers to the sensing arrangements where touch sensors are present on the external surface of robot's body and measure contact parameters through physical contact with the objects. Early works on tactile sensing focused on development of sensors exploiting various transduction methods; a large number of experimental devices and prototypes were built. Particular attention was given to the development of tactile sensors and tactile sensing arrays for object recognition [13], [16]–[23], [112]–[114]. The creation of multi-finger robotic hands in late 1980s had increased the interest in tactile sensing for robotic manipulation, and led to works utilizing tactile sensing for real-time control of arm manipulators [7], [20], [29], [113]. Researchers focussed on new applications-oriented features like mechanical flexibility and conformability, and accordingly on new designs and materials for tactile sensing. While the development of tactile sensors for robotic fingertips and hands continued, new areas such as motion planning in unstructured environment brought to the fore the idea of robot whole-body sensing. As a result, various sensitive skin design projects were undertaken in the late 1980s and 1990s [115]–[117].

Over a period of time robotics has witnessed a paradigm shift signified by the appearance of new types of robots (rehabilitation and assistive robots, bio-robots, medical robots, humanoid robots). For these devices to co-exist with humans, safety concerns and the need for precise and gentle handling of objects make it indispensable to have a continuous sensing ability distributed over the robot's entire bodies [26]. As a result, nowadays there is an increased interest in developing large area patches or whole-body tactile sensing [13], [109], [118]. The shift towards a whole-body skin has brought forth new challenging issues related to sensing data handling, data representation, data interpretation, and overall system integration. The above pattern is also visible in recently published first special journal issue on 'robotic sense of touch' by IEEE Transactions on Robotics [119].

III. MECHANICAL/PHYSICAL CONTRIBUTORS

The tactile skin's mechanical or other physical characteristics (we call them here contributors) are important for making tactile sensing structures an effective component of robotic and similar applications. These include features like skin's mechanical conformability, stretchability, spatial distribution and optimal placement of sensors, skin weight, water resistance etc. While these have always been desirable features of a robotic tactile system, only recently researchers started

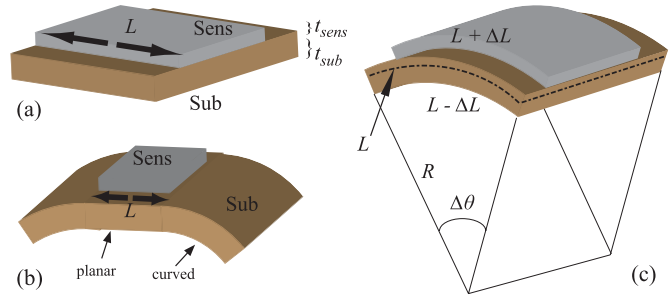


Fig. 2. Bending of a sensor structure: (a) Stiff sensor/electronic component (length - L) on top of a substrate layer; (b) Stiff sensor/electronic component on flexible substrate. The bending is restricted at locations where stiff components are present. (c) Bending of all flexible structure into a cylinder of radius R . The neutral plane (dotted lines) does not experience any change in length due to bending. The bending by $\Delta\theta$ results in change in length by ΔL .

addressing them consistently [39], [46], [107], [109]. Research towards realizing these features is reviewed in this section.

A. Flexibility and Conformability

For better integration, tactile sensing skin structures should be mechanically flexible - i.e. bendable or/and stretchable - and able to conform to arbitrarily curved surfaces of robot's body or instruments such as smart catheters. A flexible tactile skin will likely have significant impact on robotic systems like prosthetics and surgical instruments where tactile skin can impart new functionalities. The available bendable tactile sensing solutions typically use standard Printed Circuit Boards (PCB) on rubber-like substrates, or flexible PCBs as substrate. The off-the-shelf sensing and electronic components are soldered to these PCBs. The shrinking dimensions of electronic components (e.g. microprocessors) make it possible to mount devices for multiplexing, signal conditioning, and local computation in the immediate proximity of sensors. Examples of such solutions include tactile skin patches with triangular [45], [120], hexagonal [110], or the tree- or comb-shaped modules [109], [121]. Many available tactile sensing structures can bend only along one axis and hence can conform to surfaces such as a cylinder, which approximates the arm of a robot. As body parts such as shoulders require tactile skin to conform to shapes like hemisphere, triangle or hexagon shaped PCBs have been explored [45], [110], [120]. Given the usefulness of triangulation techniques in reconstructions of 3D surfaces, it should be feasible to cover the whole robot body with triangular tactile sensing modules of different sizes.

The degree of skin bendability depends on factors like thickness (t) of PCB (or other substrate), elastic modulus (Y) of substrate and the sensor layer, and the size of discrete sensing and electronic components. For instance, one of smallest commercially available capacitance-to-digital conversion chips is of the size about $4 \times 4 \text{ mm}^2$ [122], while the smallest fully integrated 3-axis accelerometer e.g. BMA250 is $2 \times 2 \times 0.95 \text{ mm}^3$, showing the potential of miniaturization. The stiffness of the chip and practical issues of interconnects restrict the overall bendability of such skin structures (Fig. 2b). Therefore, flexible PCB-based solutions are suitable

only for covering body parts with large curvature but not for parts such as fingertips of a humanoid or a surgical instrument like smart catheter [123]. The latter application call for higher bendability that can be achieved with flexible sensors and electronic components (Fig. 2(c)). Depending on the substrate/supporting material, the strain on the top surface (ϵ_{top}) of such structures (Fig. 2(c)) can be obtained as [124], [125]:

$$\epsilon_{top} = \left(\frac{t_{sens} + t_{sub}}{2R} \right) \frac{((1 + 2\eta + \chi\eta^2))}{(1 + \eta)(1 + \chi\eta)} \quad (1)$$

where $\eta = t_{sens}/t_{sub}$ and $\chi = Y_{sens}/Y_{sub}$. This means the strain on the surface top can be reduced, thus increasing the bendability, by suitably choosing the skin thickness and its material. Tactile sensing and electronics structures based on woven-like fabrics could make a good solution as they are extremely flexible, bendable, conformable and stretchable. However, present day solutions are far from this ideal scenario. Recently, significant progress has been made in flexible/bendable electronics by printing, or transferring the electronic building blocks directly, onto the flexible substrates [126]. Some examples include flexible and light-weight structures, using organic semiconductors [107], [127] and silicon nanowires based TFTs and sensors [128]–[131].

The ultra-flexible electronics is very promising but current technology is at infancy and very large scale integration will be possible in long-term. In this context, flexible or bendable chips could be of interest as they offer compact large scale integrated systems on flexible substrates, extremely low package heights, and flexible system-in-foil products. Ultra-thin chips ($\approx 50 \mu\text{m}$ thick) are realized through post processing steps like wafer thinning and then transferring the ultra-thin chips to flexible substrates [132]–[135]. It will take a few years before these novel technologies make an inroad into tactile sensing in robotics and similar applications.

B. Skin Stretchability

As mentioned above, skin stretchability is needed to cover robot body parts such as knees and elbows. There are at least three ways of attaining this feature:

- Using stretchable wires to connect sensor modules. The stretchability is introduced by suitably shaping the wires.
- Using intrinsically stretchable wires made of engineered materials to connect the sensor modules.
- Developing intrinsically stretchable transducers.

The wires/interconnects can be made stretchable by using certain shapes or patterns. For instance, the spiral-shape of telephone wire makes it possible to move the receiver away from the base. Similar techniques can be employed to obtain a stretchable tactile skin. A large-area tactile skin with expandable spiral electrodes, designed by winding the copper wires around an elastic nylon line, has been realized in [136]. Another interesting technique uses wavy wires and provides skin stretchability along a single axis [137]. The biaxial stretchability could be achieved by fabricating wavy or net-shaped conductive structures on pre-strained elastic substrate and releasing the same after fabrication. Taking advantage of

this technique, different stretchable conductors, such as metal-coated net films, wavy one-dimensional metal ribbons and two-dimensional metal membranes have been demonstrated [138]. The technique has been demonstrated at cm-scale and as such could be useful for imparting stretchability to flexible tactile skin patches for small body parts such as fingertips. It may be possible to extend the approach to larger surfaces.

Ideally, the conductors must be intrinsically stretchable like rubber, i.e. capable of regaining original state after releasing from a stretched condition, with negligible variation in the conductivity. Noteworthy efforts in this direction include using single walled carbon nano tubes (SWNT)–poly(dimethylsiloxane) (PDMS) composites, with SWNT carrying the electricity and the PDMS providing the bendability and stretchability [139], [140]. Advantages of these conductors include biaxially stretching up to 70–100% – without mechanical or electrical damage. Microfluidics has been explored in recent years as an alternative for stretchable conductors with conductive liquid confined in microfluidic channels [141]. Though promising results have been demonstrated in a laboratory set-up, it will take a few more years for the technology to be used at the scale suitable for robotic applications.

Using intrinsically stretchable transducers is another alternative for obtaining a stretchable tactile skin. A noteworthy development there is the use of electrical impedance tomography (EIT) – the technique used in medical imaging [39], [41] and geophysics [142]. An interesting feature of this approach is the presence of electrodes on the periphery of sensing patch or active sensing area only. This means that no wires crisscross the sensor material, which is typical of tactile sensing arrays based on row-column wires. Joints where EIT-based sensing structures are placed can therefore be moved with greater ease. Having stretchable tactile sensing schemes with no or minimum connecting wires in the contact region is desirable. Stretchable fabrics with knitted tactile sensing materials, tactile sensitive fabrics, fibers and wireless tactile system are other methods for obtaining stretchable tactile sensors [143]–[145].

C. Mechanical Compliance

Compliance is the skin's ability to comply with deforming forces. Compliant materials facilitate a larger contact area and hence provide greater amount of contact information. As the deformation of compliant surface propagates, a compliant tactile skin, through interpolation, can detect the tactile signal in areas not covered directly by a tactile sensors and hence has potential to extract 3-D information such as the shape of the indenting objects. Certain degree of compliance relieves the control constraints like bandwidth, update rate and delays.

Skin compliance is obtained by embedding the tactile sensors into (or covered with) elastic materials such as polydimethylsiloxane (PDMS) or by compliant sensors structures with sensors on compliant substrates (Fig. 3). In human skin receptors appear at different depths – this is a good example of the use of both types of compliance. For sensors embedded under or within the elastic substrates, the covering additionally affords some level of protection. The cover layer material must easily adhere to the transducer layer and, on the upper side,

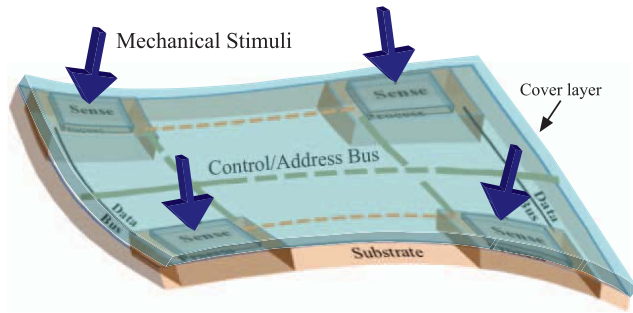


Fig. 3. Integration of transducer and electronic components, resulting in “sensing and processing at the same place.” The structure has a compliant cover layer.

must have a proper stiction coefficient with objects in order not to impair manipulation/exploration tasks.

Drawbacks from compliance due to embedding sensors in an elastic material include blurring or spatial filtering, positional uncertainty in locating contact point, and hysteresis and phase lag due to viscoelastic behavior. Finite element analyses of mechanical spatial filtering for various types and thicknesses of elastic cover materials demonstrates that this effect is a serious factor for high spatial resolution (less than 1 mm) sensors [146]. Patterning the soft cover with structures similar in size to human fingerprints or using composite materials are the alternatives to offset such blurring effects [147], [148]. Skin lines that form fingerprints are known to enhance tactile sensitivity in humans. Circumstantial evidence on how fingerprints could enhance the tactile sensitivity come from work on a biometric force sensors [149] where a soft cover is patterned with fingerprint-like parallel ridges.

D. The Number, Placement, and Distribution of Sensors

The number of tactile sensors and their distribution across the sensing surface affects the quality of measurement and eventually the effectiveness of tactile data utilization. Cues from human tactile ability tell us that the number and spatial distribution of tactile sensors vary across the body according to the functional needs, and vary relative to spatio-temporal sensitivity and pressure thresholds of body parts [8], [13], [27], [28]. Higher density of sensors translates into more precision of the motion that is generated based on tactile data. For instance, there are more receptors in human fingertips than in the palm or belly, reflecting the fact that much more tactile dexterity is required in the fingers than in the other areas. Such cues are qualitatively fine but quantitatively impractical, in the sense that the spatial distribution according to sensitivity is fine but current technology and means of data handling put limits on the feasible density of sensors. As miniaturization of sensors continues, a desired trend is to improve the tactile data processing techniques, towards extracting maximum essential information from the given sensor density. What follows is a summary of the work in this area.

As good sensing information can sometimes be obtained as with a larger number of sensors, just by strategically placing a set of sensors over the robot body. To this end, the approach adopted by Murakami *et al.* [150] to place tactile elements for grasp type recognition is noteworthy. The approach involves a

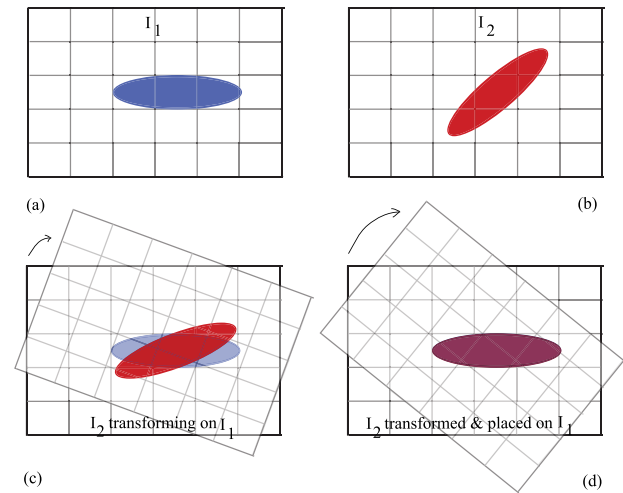


Fig. 4. Translation and rotation transforms to obtain a high resolution image using super-resolution techniques [152]. (a) The reference image I_1 ; (b) new image I_2 at a different orientation; (c) and (d) transforming the coordinate system of image I_2 to that of a reference image I_1 by rotation and translation.

human subject performing different types of grasps to obtain contact information from tactile elements installed uniformly on a glove the subject uses. A decision-making algorithm is then applied to the obtained contact information to select an effective placement of the tactile elements. The study shows that the resulting densities of sensing elements on the thumb and index finger are higher than those on the other fingers and the palm. Not only does the placement obtained by this method require fewer tactile elements (40 elements), it also achieves a recognition performance as high as for placements with many more (160) elements. Similar strategies in robotics may open optimal routes for using the tactile hardware. For instance, among multiple sensors placement configurations suitable for a task, the one resulting in minimum wiring can be opted for.

Similarly, a “super-resolution” algorithm [151], used in digital image processing to enhance the resolution of images, can be adopted to improve on the tactile sensing resolution, in case of insufficient resolution from the available tactile sensor hardware. This can happen with low cost resistive or capacitive sensors, or large size sensors, or smaller sensors placed far apart from each other. The method, depicted in Fig. 4, involves geometrically aligning into a common reference frame images taken from slightly different orientations obtained by actively touching the object [152]. Simple rotation and translation operators are then used to transform the coordinate system of an image (e.g. I_2 , Fig. 4) to that of a reference image (e.g. I_1). In case of robot sensing, transformations between multiple tactile images can be controlled using the robot pose information. Such techniques are less useful for dynamic contact due to possible variation in contact conditions during the time the images are taken. Further, the data to be processed increases by the factor of the resolution change.

The approaches such as above could open interesting avenues for using tactile data in applications. For instance, a lesser number of sensors makes it easier to handle wiring complexity, data volume, costs etc. Similarly, the lesser amount of data from low-resolution sensors will ease up the bandwidth and processing requirements of the tactile system.

E. Sensor Shapes and Layout for 3D Coverage

A well planned tactile skin structure is an important contributor towards effective utilization. For instance, carefully choosing the shape of sensor modules can help cover a given 3D surface. Example of such designs are triangle- and hexagon-shaped structures for covering large body parts of robots [45], [110], [120]. The idea originates in the triangulation technique, whereby a set of triangles is used to reconstruct 3D surfaces. A given body will thus be covered with tactile sensing modules of triangular shapes. Similarly, hexagonal shape, as in Hex-O-Skin [110], [153], [154], has been the used in vision sensing for natural triangulation. With hexagonal modules there is a natural triangulation of sensor modalities as well as a minimal variation of horizontal, vertical, and diagonal sensor distances, which helps reduce the variation of resolution in different directions, e.g. for edge detection algorithms.

Algorithms and programming tools can be used to design a tactile skin for large areas, with suitable shapes of sensor modules. A good example is that of Fuller map or the AirOcean world map [155], which is obtained by projecting the globe onto the surface of a polyhedron, unfolding and flattening of which results in a flat map of earth. Similar to tailoring human clothing, a desired 3D surface can be obtained by cutting a 2D surface and connecting/folding the pieces in a suitable manner [156]. Implementation of the above strategy for tactile skin in robotics starts with a CAD (Computer Aided Design) model of the 3D body parts of a robot, and the steps include: (a) unfolding the 3D surface and extracting the 2D representation, (b) assembling the sensing modules (irrespective of their shapes) so as to cover maximum 2D area, (c) folding the assembly of tactile sensing modules to reconstruct the 3D surface of the robot's body. The approach has been recently adopted to design the tactile skin for covering robots like Nao, Kaspar and iCub [118]. With maximum area coverage as the goal, the process can also help select suitable shapes and placement locations of tactile sensing modules for the application in hand. The above process may also help in planning the routing of the wires.

F. Wiring

Constraints on wiring are major impediment in integration and utilization of tactile skin in robotics [13], [16]. As the number of sensors increase, so is the number of wires needed to address elements, acquire and transmit the data, and power the sensors. The large number of wires bring to the fore such problems as routing of wires through the robot's joints and optimizing the use of the limited available space in the robotic shell, due to existing motors and electronic components.

Intelligent routing and addressing scheme can help reduce the number of wires. For instance, the row-column scheme for a $n \times m$ array of sensors requires $n + m$ data wires instead of $2(n \times m)$, if sensors were read/addressed individually [8]. Among approaches to sensor wiring, Nilsson [157] proposes an addressing scheme that consider sensor elements as a series of analog filters, such that they share the same transmission line and their outputs are delayed temporally much like in a Charge Coupled Device (CCD) array – so that the responses

from the sensing elements arrive sequentially at the sampling electronics. Following this technique, 16 tactile sensors require only three connectors – two for the signal carrier and one for the ground. The automated strategies used in microelectronics for designing routing wires in large-scale integration of devices can be used for tactile skin. Hybrid implementations such as interconnections on flexible PCBs/substrate have adopted some of the microelectronic routing schemes [45], [109], [110], [120], [121].

Careful sensor design is another way to reduce the number of wires. For instance, electronic components such as transistors can be used as stress sensors. Transducers can be included in the transistor structure (Fig. 3), with transducer material placed on top of gate [84], [85], [102], [103], [108] or on top of an extended gate [104]–[106] or in the source-drain path of a transistor [107], [131]. In doing so, wires needed to connect transducers with the electronics can be avoided, thus improving the sensor performance.

Wireless data communication and wireless powering of sensing hardware are also means to handle wiring related issues. Among other advantages, wireless communication will make it easier to design stretchable sensing schemes. Wireless communication for touch sensing has been attempted in [158], [159]. Wireless power transmission, as in flexible wireless power transmission sheets [160], could also be a useful step in this direction. Challenges in this area include: cross-talk among sensors over large areas, the relatively large amount of power needed, transmission delays, and common (far field) or multiple short ranging (close field) communication delays etc. There are also safety issues, especially in the context of humans working side by side with robots, as well as potential for worse reliability of wireless over the wired communication.

G. Weight of Tactile Skin

Issues related to the skin mass i.e. weight, thickness, and space needed for skin placement, are also important for effectively utilization of tactile skin. The tactile skin must be light in weight. Considering humans as an example, the adult human skin weighs about 4.1 Kg, which is about 6% of the total weight (around 70 Kg) of an adult human. Similarly, the surface area of the skin in an adult human is about 1.8 m² and the skin thickness varies between 0.5-0.9 mm [161]. Assuming these figures as reference, if a new ASIMO (Advanced Step in Innovative Mobility) [162] from Honda is to have the tactile skin, it should weigh just 3.5 Kg (the total weight of new ASIMO would be 57.5 Kg, i.e. 54 Kg for the humanoid and 3.5 Kg for the skin). Similarly, the skin for an “iCub” [163] should be about 1.5 Kg. Light weight materials and suitable designs such as honeycombs or net structured tactile sensing elements [164] are useful in controlling overall weight of skin.

IV. ELECTRONICS/ELECTRICAL CONTRIBUTORS

A general hierarchical functional and structural block diagram of a tactile sensing system is shown in Fig. 5. The diagram have been extracted and generalized from the electronic and communication infrastructure of humanoid robot tactile sensing system. The system is hierarchically divided

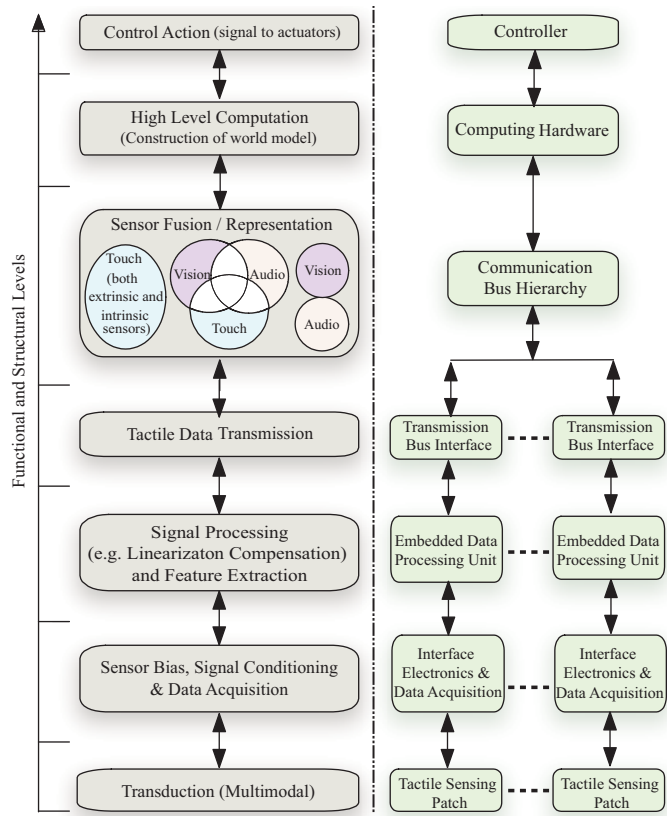


Fig. 5. General hierarchical functional (*left*) and structural (*right*) block diagrams of a tactile system. The control loop can be closed at lower levels [8], [13]. The arrows from bottom to top show the direction of tactile data transfer and from top to bottom show the addressing of sensors. Transmission channels can be heterogeneous, and can refer to different transducer types and circuit arrangements.

into blocks that reflect their functional roles and flow of sensing information and control. The levels from bottom to top depict the acquisition of sensing data, transfer of data to higher perceptual levels, and control commands to obtain the desired action. The degree of complexity vary between the hardware intensive levels at the bottom and the computation-intensive processes at the top. Below we consider some design constraints and choices for various levels shown in Fig. 5.

A. Tactile Sensors, Tactile Arrays and Tactile Patches

As basic structural building blocks of a tactile sensing system (Fig. 5), the tactile sensors and arrays influence the types of contact parameters that can be measured over an area and also the time response – which, in turn, depends on factors such as the transducer material, cover layer, number of sensors in the arrays, how the sensors are accessed and read etc. This section provides insight into the issues related to the aforementioned factors, especially the transduction method i.e. transforming the contact parameters (e.g. magnitude and direction of contact forces, distribution of force in space, mechanical stress and rate of change, strain, object proximity, temperature, etc.) into electrical signals.

The choice of the transduction method is influenced by the expected response time of the tactile sensors and arrays.

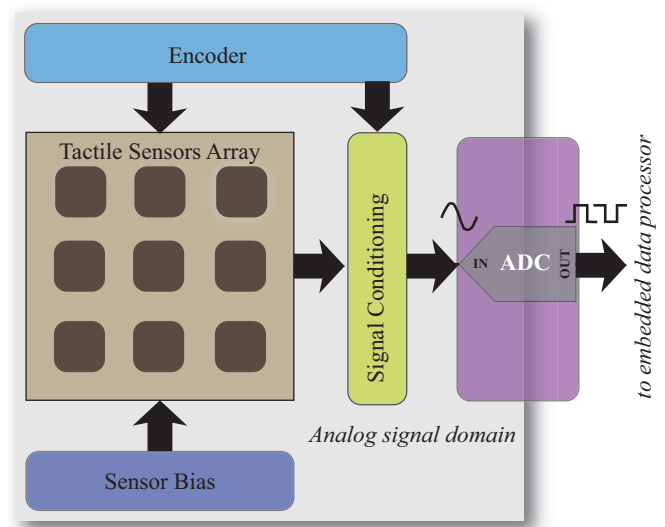


Fig. 6. General functional block diagram of the interface electronics. Each sensor is connected to a signal-conditioning channel, which can be partially or fully implemented locally for each sensor.

The maximum speed or frequency at which the transducer is capable of operating is reflected by the transducer bandwidth. High bandwidth means fast response and possibly high control bandwidth i.e. the frequency range to effectively use the tactile response in the feedback control loop of the robot (about 1 kHz). Resistive transducers achieve an overall bandwidth of some tens of Hz. Higher bandwidth e.g. more than 1 kHz, is achieved by optical [46], piezoelectric [165], [166] and capacitive [148] transducers.

Simultaneous use of more than one mode of transduction is needed many times. For example, multiple parameters such as the stress, stress rate, or temperature can be measured with a set of sensors that are based on different transduction methods, like capacitive (for strain measurement) and piezo/pyroelectric (for stress or temperature measurement). An ideal scenario would demand a tactile sensor to use the same mode of transduction to measure multiple contact parameters - in other words, a transducer that is “multimodal”. Advantages of multimodal transducers also include reduced sensing hardware.

B. Interface Electronics

A general functional block diagram of interface electronics is shown in Fig. 6, with blocks indicating sensor bias, signal conditioning, and data acquisition [56], [64], [84], [167], [168].

1) *Sensors Bias*: From the electronics viewpoint, tactile sensors are classified into two categories: (a) passive sensors (e.g. capacitive, resistive), and (b) sensors based on active devices (e.g. Field Effect Transistors (FET) [84], [169]). The bias circuit is needed to fix the operating point and to get rid of the switch-on time of the tactile sensor in case it involves an active device. The bias circuit is usually comprised of a current source which fixes the DC channel current of the FET and keeps it in active mode in the steady state conditions i.e. when no input stimulus is applied.

2) *Signal Conditioning*: This block implements a set of circuit level functions such as low noise amplification, input/output impedance adaptation, setting the reference DC values (e.g. signal ground), low-pass or band-pass filtering (e.g. anti-aliasing low-pass filter) [84], [167], [168]. The output of the signal conditioning circuit is input to analog to digital (A/D) converter. Usually a dedicated signal conditioning channel is needed for each sensor element in the array with a single A/D converter. Signal conditioning channels (one for each sensor element of the array) are time multiplexed via an analog multiplexer at the input of the A/D converter.

3) *Data Acquisition*: Data acquisition involves addressing the signal conditioning channel and digitization (i.e. analog to digital conversion) of the analog input. The tactile sensing control can query and read tactile data from any sensor in the system. The ability to query individual tactile sensors is helpful e.g. for diagnostics and calibration. Issues like wiring complexity also influence the interface electronics, in particular for large skin area arrays. The addressing can be serial and fixed [46] to decrease complexity at the expenses of an increase of array scanning rate.

An alternative arrangement for signal conditioning and data acquisition is to translate the sensed signal into a frequency value, which is subsequently digitized and acquired via a digital counter. The advantage of this approach is the robustness of the sensor output signal with respect to noise and disturbances. Wiring is minimized as the sensor output is a single signal/wire. On the other hand, the acquisition time can be very long as it depends on the oscillating frequency and on the sensing resolution.

C. Embedded Local Data Processing Unit

Difficulties with the acquisition and processing of large data from many tactile sensors has been cited as one reason for slow penetration of tactile sensing in robotics vis-a-vis other sense modalities [24]. Pre-processing of sensing data close to the sensor location, i.e. before transmitting to higher levels of the tactile sensing system, can significantly reduce the amount of information between the sensors, computing hardware and controllers. Such distributed architecture can help reduce the throughput of the robot processing unit and free up the “robot brain” for more intelligent work. It can also help scale up the system to practically any large number of sensors.

The embedded local signal pre-processing include computations such as scaling (amplification), segregation of signals from different kinds of sensors (e.g. force, temperature), linearization, and temperature compensation. Other more complex real time tasks are often required as reported in [167], [170]–[172]. The role of a slip sensor [168] in an anthropomorphic robot hand is to detect object slip and provide information to a controller to allow it to adjust the grip force needed to securely hold an object. The embedded local processing could reduce the cognitive load of a robot by not having to visually assess the stability of the object, as well as giving it the confidence of not applying unnecessarily excessive grip forces. Local processing could also be useful in instances where a set of complex features (e.g. minimum/maximum,

spatial resolution, duration etc.) are extracted [170] and input to the classifier of the contact touch modality. In such cases the inverse problem is needed to be solved in real time as well to compute the contact image at each scan iteration.

A complete sensing system (hardware + software) for an anthropomorphic robotic hand which closely matches the low levels structure (up to the Tactile Data Transmission Level) reported in Fig. 5, is presented in [173]. The control loop in this case can be closed at the hand hardware level. The system can be interfaced to different transmission protocols (e.g. CAN, USB, RS232) depending on the communication infrastructure of the robot. Signal processing tasks and classification (e.g. slip detection, contact, etc.) are implemented by the embedded hardware thus decreasing the computational burden on higher computational levels.

The issues and challenges for the implementation of large scale robotic skin at structural level are assessed in [120]. Some examples of the implementation of a sensing skin based on capacitive transducers on different robotic platforms (e.g. iCub, Nao, Kaspar) are presented and experimentally evaluated. Following the proposed modular approach, the skin modules can be arranged in such a way to cover big areas of the robot body with reduced wiring. The skin data are used to control the grasping of objects.

A different approach is presented in [110], where the basic sensing module (HEX-o-SKIN) is implemented by hexagonal PCB which hosts a microcontroller for data pre-processing (e.g. clustering, feature extraction, etc.) and transmission interface, and a bunch of custom and off-the-shelf sensors (e.g. proximity, pressure, vibration, temperature, etc.). HEX-o-SKIN can be arranged in a modular way to configure skin patches hierarchically interfaced to the robot backbone. The system is robust in terms of hardware transmission faults (i.e. broken connections) as it implements redundant network connections and automatic routing procedures to bypass the fault. A prototype network has been integrated on a KUKA lightweight robotic arm. The work has been advanced to prototypes having 15 and 74 modules, which shows the maturity of technology [148]. Experimental results demonstrate that based on the tactile feedback, the robot is able to react to the lightest touch, multiple-touch inputs, to balance a cup on a plate at the end-effector, to react to impacts and air drafts.

The System on Chip (SOC) or System in Package (SIP) approach benefit from the local embedded implementation of tactile data pre-processing algorithms. SOC/SIP can be seen as a tactile analog of CMOS optical arrays/imagers, which have significantly contributed in bringing vision sensing to applications. The approach has been used in work on smart fabric [174], MEMS based cantilever force sensors [175], and smart vision [176].

D. Communication and Transmission

Dedicated strategies are needed to transmit, to the computation and control units, the data collected by the tactile sensor arrays spread over the whole robot body. A general computational framework which aims at efficiently managing the communication and electronic infrastructure

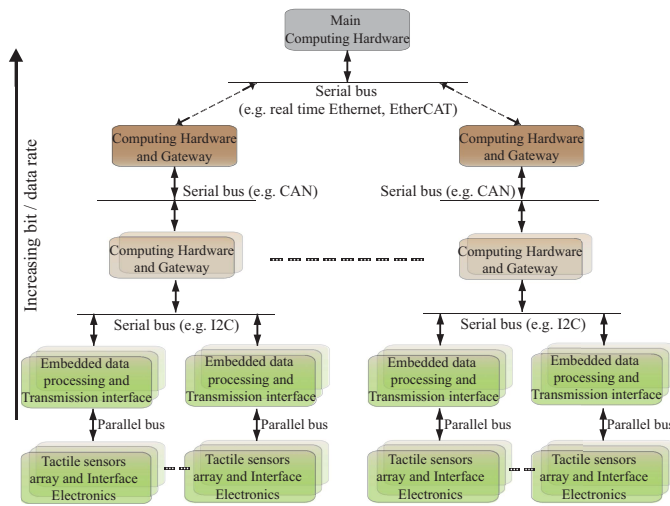


Fig. 7. Hierarchical communication architecture in the tactile sensing system.

is presented in [177]. The approach applies to networks of “sensorial material” i.e. functional/structural material patches with embedded dedicated electronics which easily fit the computing/communication structure arrangement of a sensitive electronics skin. Nonetheless, a proper communication infrastructure is anyway needed for the effective implementation.

In this context, the hierarchical architecture of the communication bus and the embedded local data processing can be exploited. Along with what happens in the human tactile system, where sensed data are progressively processed and salient features are extracted along the path from the skin to the central processing unit, the brain – similarly, along the path from periphery to the robot controller, local buses are linked at connection points (gateways) where data from periphery are collected, processed for sensor fusion, feature extraction etc. and forwarded to higher levels. Gateways are themselves arranged in a hierarchical network topology up to the central control unit. The communication bus in this topology is star-like with a hierarchal arrangement. The gateway from one layer to the other has local computing resources to implement the appropriate data processing algorithms (see Fig. 7). Only salient features are transferred to higher levels.

Going from lower to higher levels (i.e. from skin towards controller), the protocols are different. Serial communication buses are used to decrease the amount of wiring. Namely, the bandwidth of the bus increases in order to accommodate an increasing amount of data; the protocol complexity and performance increase as well. At lower levels high speed, lower connectivity and short distance wiring buses are preferred (e.g. I2C up to 400 kbit/s [178], see Fig. 7). Moving up in the hierarchy, more complex protocols and longer wiring buses are preferred (e.g. CAN 1 Mbit/s, Flexray 10 Mbit/s, real-time Ethernet 100 Mbit/s). A similar arrangement is presented in [120], where a skin patch hosts an embedded processing hardware and communication interface (I2C). Data are sent to a microcontroller board (capable of managing up to 12 skin patches) to implement the communication interface with an efficient protocol i.e. CAN bus or EtherCAT. The embedded

hardware measures 12 taxels independently (16-bit data) at 50 Hz or an average of them at about 500 Hz. The middleware supporting the skin systems in real-time and the performance for a whole body robot skin are demonstrated in [179].

A thorough assessment of a networking infrastructure based on 100 Mbit real-time extensions of the Ethernet protocol, aimed is presented in [180]. EtherCAT and PowerLink are considered and compared on the whole robot body network. Even though both protocols present very high bandwidth, EtherCAT performance are superior to those of Powerlink. Focusing on at least 160 microcontrollers (one for each skin module) and assuming 192 byte data packets, realistic cycle of times EtherCAT are in the order of 1.9–3.0 ms, which are time responses compatible with reactive robot behaviours.

A completely different approach is presented in [110], [148], where the skin modules are organized in a mesh network with “multi-hop” communication. The network does not present a hierarchical arrangement as each module has 4 ports providing 4 UART connections with the n neighboring modules, another skin patch or the robotic backbone. The update rate with eight tactile modules (1 kHz) is quite impressive and it allows to close the control loop even at higher processing levels.

Wireless data transmission is another alternative, which also helps reduce the wiring complexity. In the work by Shinoda *et al.* [158], [159], tiny wireless sensing elements are embedded in an elastic skin with an inductive base coil that provides power and signal transmission. Each sensing element is a tuned stress sensitive resonator with a distinct resonant frequency. Wireless networks, which are based on radio [181] or optical transmission [182], suffer from low bandwidth and interference among (many) sensing elements placed close to each other, require a complex supporting structure and a relatively high power consumption. There are also safety issues when robots are to work near humans, as well as an issue of reliability of wireless over wired data communication.

E. Other Electronic/Electrical Requirements

The power consumed by an individual tactile element (taxel) should be low since it will be multiplied by the total number of sensors which, in the perspective of a whole body skin, could be very high. Results in [120] present a power consumption of about 5 W/m^2 i.e. for some hundreds of taxels. The power consumption in [110], [148] is about 130 mA at 3.3V per tactile module which is fairly comparable to that of [120]. Optoelectronic transducers [183] present a fairly high power consumption e.g. 200 mA per taxel. Such drawbacks along with the system complexity prevent their adoption in large area skin applications. Power consumption is especially an acute cause of concern when sensors are used in an autonomous robot that relies on battery power.

The sensor array and interface electronics are generally powered at different supply voltages because their implementation technologies are usually different. In robots, the power supply to peripheral circuits, such as the interface electronics for the skin, can be taken from a dedicated cable (with power supply voltage and ground reference). If possible, the power supply can also be taken from the communication bus wires to

keep the wiring simpler. For instance, the +5 V supply can be taken from the CAN bus wire, even if it is noisy, and cannot be used as reference for analog circuit functions. If other supply voltage values (e.g. +3.3 V or +1.8 V) are needed (e.g. for a microcontroller, DSP, or Application Specific Integrated Circuit), dedicated voltage regulators must be used. Besides increasing the physical size, the power consumption of power circuits also worsens the system energy budget. Local power sources based on energy harvesting systems [184], [185] are viable solutions for distributed and local supply power sources without the need of a periodic recharge.

The power supply and reference (ground) potentials maybe taken from long wires, as the reference potential values vary over the system, due to voltage drops along the wiring cables and due to varied power consumption in different units. This issue can be tackled effectively with a local power supply source e.g. fuel cells or energy harvesting circuits, or with steps like buffering of energy in the local voltage regulation circuit by passively distribution of power [110].

V. DATA MANAGEMENT - SOFTWARES AND ALGORITHMS

A. Feature Extraction and Data Handling

Similar to the human tactile system, data transmission in the robot skin can be done “in parallel” with features extraction, by exploiting a communication bus hierarchy. Considering tactile events rather than the raw signals will influence the way the control algorithms and classifiers are used. This may help reduce the amount of transmitted tactile data.

Various computational techniques have been used to extract information from tactile data and to address complex tasks like grasp stability and safe grasping. For instance, tactile information from the contact between a gripper and the object have been used to obtain the pose of the object via particle filter technique [186]. Recognition of contact material using Support Vector Machine (SVM), Regularized Least Square (RLS) and Regularized Extreme Learning Machine (RELM) is reported in [187]. Similarly, neural network algorithms are used to infer from tactile contact information various object surface features (edge, flat, cylindrical, sphere) [188]. Tactile information was shown useful in determining deformation properties of objects and “open/closed” and other parameters of bottles [189]. Machine learning tools have been used for surface recognition and categorization based on the data obtained by letting a robot scratch different surfaces [190]. Self-organizing maps (SOMs) have been employed to extract various features from sensory data [191]. Based on information extracted from contacts, objects are distinguished according to their shape, size, texture, and hardness. Similarly, Bayes trees have been used to distinguish different materials based on information on their surface texture [192].

B. Data Representation and Sensor Fusion

The computational techniques developed for computer vision have also been adopted in robotic tactile sensing. Since robot tactile sensors are distributed over the 3D robot body, information from those sensors, unlike with cameras, is geometrically more complex and ill-defined. This affects the

processing of tactile data at the higher computational levels (Fig. 5). The changing positions of tactile sensors in space with the robot motion also makes processing of tactile data more labor intensive and error prone (especially in the case of whole body tactile sensing). Such processing is fundamental to the development of more complex cognitive behaviors, such as quick response to sudden stimuli or compliant human-robot interactions. Internal models must be developed to represent the value of tactile information for various robot body parts. In humans, somatotopic internal maps are used for these purposes. Similar arrangements are called for in robotics. Inspired by the human somatosensory system, some researchers make use of models based on the notion of “somatic” alphabet, words and sentences [193], and artificial brain maps [194]. A comprehensive approach here is yet to evolve.

A method similar to human somatotopic maps is employed by Stiehl *et al.* [193] to create receptive fields from a population of sensors. Their algorithms, fashioned over the brain processing methods, attempt to determine the centroid of an object, as it moves relative to a group of sensors. The algorithms can also be used to determine object properties such as texture, curvature, and orientation. The approach does not take into account mapping between the taxels in 3D space and their representation in the 2D artificial somatosensory maps – which would be useful in contact control by enabling the robot to obtain accurate tactile data no matter what is the posture. Cannata *et al.* [194] consider such mapping by modeling the robot skin as a discrete 3D mesh of triangles. Surface parametrization techniques [195], [196] are then used to obtain a piecewise linear mapping between a given 3D surface (skin meshes) and an isomorphic discrete 2D surface while preserving the intrinsic properties (taxels locations, displacements, varying density and resolution etc.). Acquiring the 3D surface of robotic body parts based on tactile exploration requires a large number of interactions and additional constraints to resolve ambiguities [197], a calibrated kinematic and/or CAD model and additional sensors e.g. force torque sensors [198]. A method to instantaneously reconstruct the 3D shape of cloth like sheets has been introduced by Hoshi *et al.* [199] utilizing a fixed grid of gravity and magnetic sensors. The results shown in Fig. 8, extend the above method to support arbitrary skin patches of a cellular artificial skin, featuring only a single accelerometer per cell [153]. Representing the 3D skin as a connectivity graph of neighboring cells, the system estimates the local, relative homogeneous transformations solely from the static size of the unit cell, dynamic knowledge on the cell-to-cell connections and a set of gravity vector measurements in different poses. Applying graph algorithms it is then possible to efficiently calculate accurate (< 5% loop error) relative transformations between two arbitrary cells e.g. the relative position and orientation of every cell in a single skin patch. Knowing the shape of the comparably rigid body parts, the challenge is to map those tactile body schema models into an existing kinematic representation or build and calibrate a new one from this starting point - offering interesting research for fusion of visual and tactile information for calibration [200]. One issue with such models is the amount of computational resources needed and the constraints that it places on hardware.

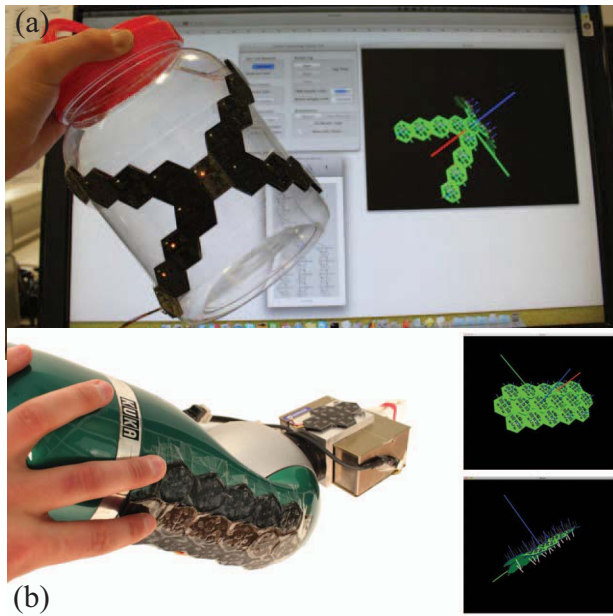


Fig. 8. 3D reconstruction of a skin patch placed on (a) a container, and (b) a KUKA LWR arm [153].

Constructing a realistic world model may require the simultaneous use of different sensing modalities (e.g. tactile, vision, kinesthetic). For instance, consider a task of differentiating between two objects that requires visual information (say color and shape) as well as tactile information (say softness, texture, and temperature). Sensor fusion is not a new area in robotics. For instance, in a typical task of simultaneous localization and mapping (SLAM) for a mobile service robot, inputs from different sensors (laser scan, ultrasonic, vision) are combined to drive the prediction algorithms. Humans also combine sensory data from different sensing modalities to interact with the environment [201]. Sensor fusion may be particularly important for monitoring contacts, e.g., for selecting impedance parameters or for monitoring some control points on the robot body that must be kept away from the nearby humans. Suitable approaches here include artificial intelligence (AI) techniques (fuzzy sets, neural networks). Using multimodal sensory inputs (proprioceptive and load-based tactile information), Platt *et al.* [202] have proposed an approach to localize the features embedded in flexible materials during robot manipulation. It is shown that in this process the proprioceptive and tactile data contain complementary information. Kroemer *et al.* [203] use a weak pairing of tactile and visual information in a machine learning approach that helps infer lower-dimensional representation of tactile data and classify materials.

Combining more than one sensor modalities may impose compatibility constraints. For example in humanoid robots sensing information may come from a combination of extrinsic and intrinsic touch sensors, vision sensors and audio sensors [204], [205]. This fact may impose constraint on the tactile sensor density, resolution, and location on robot body.

VI. OTHER CONTRIBUTORS

Besides aforementioned requirements and constraints, the developer of a tactile sensing system faces issues like

manufacturability, fault tolerance, system reliability and maintainability, and cost. Some of these are discussed in this section.

A. Modular Approach

Modularity is an established technique for organizing and simplifying a complex system. From elementary (flashlight batteries) to complex (cells in a live organism) systems, modularity has a record of success that is hard to challenge. Benefits of modularity bring forth for a tactile or other sensing system include: scalability, i.e. ability to scale and grow; flexibility, i.e. simple duplication; simplicity, i.e. of specialized functional of modules; portability, i.e. rapid adaptation to the requirements; reliability, i.e. fault tolerance etc.; and, short development time. The robustness of a robot depends much on its sensors and actuators with which it interacts with the environment. Due to the cost and other considerations, today's robots are designed with little redundancy. A big role is played by miniaturization and modularization [25], [46], [115].

B. Fault Tolerance, Graceful Degradation, Reliability

As the robot tactile sensing system may comprise thousands of sensors and associated electronic components, its reliability becomes a serious issue. To make a system more reliable, it must include means for fault tolerance, graceful degradation, self-diagnostics, and self-healing [25].

Fault tolerance is the ability of a tactile sensing system to preserve its overall functionality over loss of some sensors. In a system with large number of sensors some of those sensors will inevitably go bad, and some will go bad earlier than others – hence the fault tolerance issue. At the materials level such strategies as by Yuan *et al.* [206], for fault-tolerant soft electronics, are useful. The high-strain-capable carbon-nanotube electrodes that selectively combust or “self-clear” are used in localized short circuits to maintain constant operation of the dielectric elastomer actuators. At hardware level, fault tolerance is achieved by conventional means – a high quality of hardware components, component redundancy, and careful design of communication and routing schemes.

A less conventional means and an essential resource to achieve fault tolerance is appropriate processing of sensor data - that is, using suitable algorithms and software. As a simple example, when the system determines that a given sensor has died or cannot be trusted (e.g. due to hard-to-measure internal electronic drift), a sophisticated algorithm can replace the output of that sensor with interpolated values of the neighboring sensors. That is, without interrupting the robot operation, the missing sensor will be “replaced” by a “virtual sensor” whose output is not as good as that of the original sensor but acceptable. That is, we'll get a somewhat degraded performance. Such and more sophisticated algorithms [207] help identify, isolate, and mitigate faulty devices, prolonging the system's overall functional life.

Self-diagnostics refers to the continuous or regular checking of the system's components, such as sensors and associated electronics, and informing the controller about detected deviations from the normal behavior. Self-healing implies an

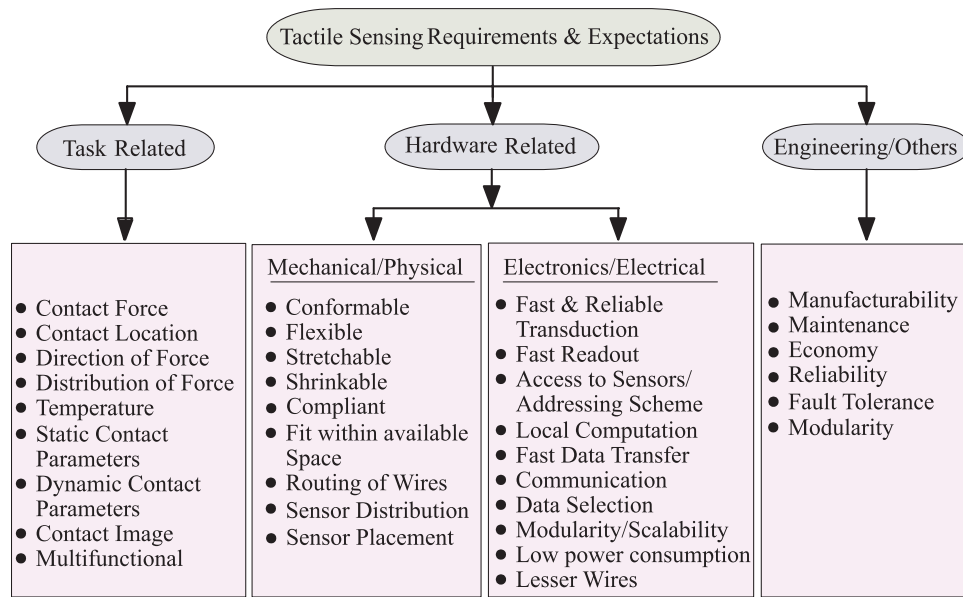


Fig. 9. Summary of tactile sensing system requirements and expectations.

automatic repair of the failed hardware – a feature common in biological systems. As the tactile skin is physically in contact with objects during robot tasks, it is subjected to constant damage. A flexible tactile skin able to measure contact parameters and yet self-heal will be very useful for soft robotics and prosthetics. It is challenging to combine such properties. For instance, most of the self-healing materials focus on designing unique-stimulus responsive system for the restoration of mechanical properties [208]–[210] – ignoring the electrical properties e.g. conductivity. In recent years, there has been intense research to overcome this limitation using composite materials [211]–[214]. The composite materials, comprising of supramolecular organic polymers with embedded nickel nanostructured microparticles, have been recently shown to possess mechanical and electrical self-healing features suitable for electronic skin applications [215]. The research on materials with self-healing properties is required to be strengthened with new strategies e.g. to overcome partial loss of connections or electronic components.

The graceful degradation refers here to the sensing system capacity to direct sensors in the vicinity of a broken sensor to take over its job. The price to be paid for such “replacement” is degradation of system performance. Namely, the distance between well functioning sensors will now effectively increase resulting e.g. in poorer system resolution. Nonetheless, the system will still function.

C. Durability, Manufacturability, Maintenance, and Cost

Good manufacturability, maintenance and low cost are other desirable items in an effective tactile sensing system. The development process itself is therefore a key issue, involving selection of materials and components, development of methods and procedures, and other processes. Computer-aided design (CAD) and computer-aided engineering (CAE) tools can be used to compare design alternatives as to their properties and cost effectiveness. Promising and versatile ink-

jet [216] and screen printing [217] technologies have been used to implement diverse tactile sensors and multimodal patches at low cost on flexible substrates. Similarly, the printing of electronic devices (e.g. Field Effect Transistor) will pave the way for implementing interface electronics on a flexible substrate [126]. Thus, printable electronics techniques, currently being investigated for large area electronics, will be useful for development of cost effective tactile skin also.

VII. CONCLUSION

Tactile skin, as large array of sensors and electronic components embedded in flexible, bendable and stretchable substrates, will endow robots and biomedical instruments the ability to sense their surroundings. It will revolutionize important areas of service industry, make crucial contributions to human prosthetics, and augment human sensing when fashioned into clothing. The state of the art in the areas that are basic to development of tactile skin shows that a large spectrum of highly efficient sensing devices are feasible. Existing prototypes are often clumsy, have low resolution, accuracy and reliability, which makes it difficult to effectively utilize them in many robotic tasks. The designing of a tactile skin and its integration in the robotic system is a result of many structural, computational and electrical/electronics hardware trade-offs – sensor modalities, robot controller embedding electronics, distributed computing power, communication networking, wiring, power consumption, robustness, manufacturability, maintainability, and so on. Some of the features needed in tactile skin for its effective utilization are summarized in Fig. 9. Serious research challenges elaborated in this paper have to be considered before tactile skin can become a ubiquitous presence in new generation robotic devices. We hope the readers will view this paper as our first effort to map out the new directions and challenges in utilizing tactile skin, and as an invitation to join in the exploration.

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Ravinder S. Dahiya (S'05–M'09–SM'12) is a Senior Lecturer with the Electronics and Nanoscale Engineering Research Division, University of Glasgow, Glasgow, U.K. He received the B.Tech (Hons.) degree in electrical engineering from Kurukshetra University, Haryana, India, in 1999, the M.Tech. degree from the Indian Institute of Technology Delhi, Delhi, India, in 2001, and the Ph.D. degree in humanoid technologies from the Italian Institute of Technology, Genoa, Italy, and University of Genoa, Genoa, in 2009. From 2001 to 2008, he was a Lecturer with the Netaji Subhas Institute of Technology Delhi, Delhi. From 2008 to 2010, he was Post-Doctoral Researcher with the Italian Institute of Technology, Genoa. From 2010 to 2013, he was a Researcher (Marie Curie Fellow) with Fondazione Bruno Kessler, Trento, Italy. From April to July 2013, he was a Visiting Researcher with the Centre for Large Area Electronics, Electrical Engineering Department, University of Cambridge, Cambridge, U.K. His current research interests include flexible electronics, tactile sensing, and electronic skin. He is the author of *Robotic Tactile Sensing – Technologies and System* (Springer). He has published more than 70 scientific papers and holds one patent. He has worked on many international projects and is currently coordinating a European Commission-funded collaborative project. He is an Associate Editor of the IEEE TRANSACTIONS ON ROBOTICS and a Guest Editor of the IEEE SENSORS JOURNAL Special Issue on Flexible Sensor and Sensing Systems. He was the Guest Editor of the IEEE TRANSACTIONS ON ROBOTICS Special Issue on Robotic Sense of Touch. He is an AdCom member of the IEEE Sensor Council as a representative of the IEEE Robotics and Automation Society. He was awarded with the University Gold Medal in 1999. He received Best Paper Awards on two occasions: in IEEE-sponsored national and international conferences, in 1998 and 2007, respectively.



Philipp Mittendorfer received the bachelor's degree in electrical engineering from Technische Universität München, Munich, Germany, in 2008, and the Diploma degree in 2009. Since December 2009, he has been with the Tactile Sensing Group with Prof. Gordon Cheng, Institute for Cognitive Systems, Technische Universität München. His current research interests include methods to self-organize intelligent surface sensor networks and their practical implementation as multimodal artificial sensor skins into robotic systems.



Maurizio Valle (M'01) received the master's degree in electronic engineering and the Ph.D. degree in electronic engineering and computer science (curriculum microelectronics) from the University of Genoa, Genoa, Italy, in 1985 and 1990, respectively. In 1992, he joined the Department of Biophysical and Electronic Engineering (DIBE), University of Genoa, as an Assistant Professor. Since January 2007, he has been an Associate Professor of electronic engineering at DIBE. He has been in charge of the Microelectronics Section, DIBE Research Laboratory, since 2002. He has been and is in charge of many research projects funded on the local, national, and European levels and by Italian and foreign companies in the field of electronic and microelectronic systems. His current research interests include embedded electronic and microelectronic systems, integrated sensors circuit interfaces, wireless sensors networks, and tactile sensing systems for robots. He is the author or co-author of more than 160 scientific papers published in national and international conference proceedings and journals.



Gordon Cheng (SM'06) is the Professor and Chair of Cognitive Systems, and founding Director of the Institute for Cognitive Systems, Technische Universität München, Munich, Germany. He was the Head of the Department of Humanoid Robotics and Computational Neuroscience, ATR Computational Neuroscience Laboratories, Kyoto, Japan, from 2002 to 2008. He was the Group Leader for the JST International Cooperative Research Project, Computational Brain, from 2004 to 2008. He was designated a Project Leader from 2007 to 2008 for the

National Institute of Information and Communications Technology of Japan. He has held visiting professorships worldwide in multidisciplinary fields comprising mechatronics in France, neuroengineering in Brazil, and computer science in the USA. He held fellowships from the Center of Excellence and the Science and Technology Agency of Japan. Both of these fellowships were taken at the Humanoid Interaction Laboratory, Intelligent Systems Division at the Electrotechnical Laboratory, Japan. He received the Ph.D. degree in systems engineering from the Department of Systems Engineering, The Australian National University, in 2001, and the bachelor's and master's degrees in computer science from the University of Wollongong, Wollongong, Australia, in 1991 and 1993, respectively. He was the Managing Director of the company G.T.I. Computing in Australia. His current research interests include humanoid robotics, cognitive systems, brain machine interfaces, biomimetic of human vision, human-robot interaction, active vision, and mobile robot navigation. He is the co-inventor of approximately 15 patents and has co-authored approximately 180 technical publications, proceedings, editorials, and book chapters.

He is a Senior Member of the IEEE Robotics and Automation and Computer Society. He is on the editorial board of the *International Journal of Humanoid Robotics*.



Vladimir J. Lumelsky (M'80–SM'83–LF'94) is a Professor Emeritus with the University of Wisconsin-Madison, the University of Maryland, and the an Independent Consultant. He received the Ph.D. degree in applied math from the Institute of Control Sciences, Russian National Academy of Sciences, Moscow, Russia. He has held research and faculty positions with Ford Motor Research Labs, General Electric Research Lab, Yale University, New Haven, CT, USA, University of Wisconsin-Madison, Madison, WI, USA, University of Maryland, College

Park, MD, USA, and NASA-Goddard Space Center. He held visiting positions with the Tokyo Institute of Science, Tokyo, Japan, Weizmann Institute, Israel, USA National Science Foundation, and the USA-Antarctica South Pole Station. He has served on Editorial Boards of the IEEE TRANSACTIONS ON ROBOTICS AND AUTOMATION, IEEE SENSORS JOURNAL as a Founding Editor-in-Chief, and other journals, on governing bodies and committees of IEEE (including IEEE Fellow Committee), IEEE Robotics and Automation Society, IEEE Sensors Council (currently as President), as a Chair and Co-Chair of major international conferences, and as a Guest Editor for special journal issues. He has served as a consultant to NSF, DARPA, and the European Commission, and as a litigation technical expert. He has authored over 200 publications (books, journal papers, conferences, reports). He is a member of ACM and SME.